

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 8_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Write a program to validate the email address and display suitable exceptions if there is any mistake.

Create 3 custom exception classes as below

DotException AtTheRateException DomainException

A typical email address should have a "." character, and a "@" character, and also the domain name should be valid. Valid domain names for practice be 'in', 'com', 'net', or 'biz'.

Display Invalid Dot usage, Invalid @ usage, or Invalid Domain message based on email id.

Get the email address from the user, validate the email by checking the

above-mentioned criteria, and print the validity status of the input email address.

Input Format

The first line of input contains the email to be validated.

Output Format

The output prints a Valid email address or an Invalid email address along with the suitable exception

If email ends with . or contains not exactly one . after @, it throws:

DotException: Invalid Dot usage

Invalid email address

If @ appears not exactly once, it throws:

AtTheRateException: Invalid @ usage

Invalid email address

If the part after the last dot is not among accepted domains:

DomainException: Invalid Domain

Invalid email address

If all conditions satisfied then print:

Valid email address

Refer to the sample input and output for format specifications.

Sample Test Case

Input: sample@gmail.com

Output: Valid email address

Answer

// You are using Java

```
import java.util.*;
```

```
class DotException extends Exception{
    public DotException(String message){
        super(message);
    }
}
```

```
class AtTheRateException extends Exception{
    public AtTheRateException(String message){
        super(message);
    }
}
```

```
class DomainException extends Exception{
    public DomainException(String message){
        super(message);
    }
}
```

```
class ValidEmail{
    public static void main(String args[]) throws
    DotException,AtTheRateException,DomainException{
        Scanner sc = new Scanner(System.in);
        String mail=sc.nextLine();
        Boolean result=true;
        try{
            int dotCount=0;
            int atCount=0;
            for(char c : mail.toCharArray()){
                if(c=='.'){
```

```

        dotCount++;
    }
    else if(c=='@'){
        atCount++;
    }
}
if(dotCount!=1 && mail.indexOf("@")<mail.indexOf(".")){
    result=false;
    throw new DotException("Invalid Dot usage");
}
if(atCount!=1){
    result=false;
    throw new AtTheRateException("Invalid @ usage");
}
if(!(mail.endsWith(".com") || mail.endsWith(".in") || mail.endsWith(".net") ||
mail.endsWith(".biz"))){
    result=false;
    throw new DomainException("Invalid Domain ");
}
}
}
catch (DotException e){
    System.out.println(e);
}
catch (AtTheRateException e){
    System.out.println(e);
}
catch (DomainException e){
    System.out.println(e);
}
}
if(result){
    System.out.println("Valid email address");
}
else{
    System.out.println("Invalid email address");
}
}
}
}

```

Status : Correct

Marks : 10/10